

Raunak Kumar

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SUMMARY

B.Tech Artificial Intelligence student at SVNIT with strong foundations in **data structures, algorithms, and systems-level software design**. Experienced in building production-grade, **real-time and concurrent systems** in **C++, Go, and TypeScript** – including a sharded, low-latency WebSocket game server – alongside applied machine learning research in Python. Active competitive programmer (**Codeforces Specialist, rating 1554**) with a demonstrated ability to reason about performance, concurrency, and correctness under real engineering constraints, developed through academic research, competitive programming, and large-scale engineering projects.

TECHNICAL PROJECTS

Ongoing Research

Class-Imbalance-Aware Brain MRI Representation Learning

PyTorch, MedSAM

- Investigating **4-class brain MRI tumor classification under class imbalance** on **2,870 scans** using **MedSAM** embeddings in Python and PyTorch; built a comparative pipeline spanning frozen baselines, fine-tuning, VAE latent modeling, SimCLR, SupCon, and hybrid SupCon-VAE, evaluated via accuracy, weighted F1, silhouette score, and UMAP analysis.
- Identified a **48-D SupCon-VAE latent space** as the strongest configuration (**58.4% accuracy, 0.543 weighted F1**); documented failure modes and defined CVAE-style next steps for further investigation.

Selected Projects

[Chessing](#) | Full-Stack Real-Time Multiplayer Chess Platform

Go, NestJS, Next.js, Socket.IO, Redis

- Architected a **three-service distributed system** (Next.js frontend, NestJS backend, standalone Go game server) supporting concurrent live multiplayer chess, time-controlled play, matchmaking, and a full metagame layer (factions, tournaments, wagering); built a **256-shard, lock-free concurrent WebSocket hub** in Go using FNV-1a hashing for high-concurrency real-time state management, with **400ms clock lag compensation**.
- Implemented a full **Glicko-1 rating engine** (rating deviation tracking, per-pool ratings across five time controls) and **Redis Sorted-Set based matchmaking** with dynamic ELO-range expansion and Redis pub/sub for cross-instance game synchronization; integrated Stockfish and Maia (Leela Chess Zero) engine pools for move validation, AI opponents, and post-game analysis.
- Engineered a **scalable backend** with PostgreSQL for persistent game records and rating history, Redis for real-time lobby/session state, and a dedicated **Go-based load-testing harness**; containerized all services with Docker Compose (tuned to a 150K file-descriptor limit for high WebSocket concurrency); secured access with **JWT authentication** and structured the NestJS layer (23 modules) with modular guards, interceptors, and DTOs.

Sprint Serve | Full-Stack Restaurant Operations Platform

[GitHub \(private\)](#)

- Designed a **scalable full-stack restaurant operating system** with a TypeScript + Express backend, Next.js admin dashboard, and guest-facing PWA supporting QR-based ordering, shared dining sessions, reservations, billing, and analytics; applied modular software design across independent service layers for maintainability at scale.
- Engineered **concurrent real-time workflows** via Socket.IO across guest, waiter, kitchen, and admin interfaces; designed a normalized Prisma + PostgreSQL schema with **transactional consistency** under concurrent writes, order-item price snapshots, and session-close flows guaranteeing data integrity.
- Containerized with **Docker Compose, Redis, and Nginx**; implemented passwordless OTP + JWT authentication, geo-verification, Helmet, rate limiting, structured logging, and Sentry monitoring for production reliability and debugging.

Joy Juncture | Full-Stack Experience Commerce Platform

[GitHub](#)

- Built a **multi-module full-stack platform** for game commerce, event booking, and experiential services using **Next.js 16, React 19, TypeScript, Prisma, PostgreSQL, Tailwind CSS, and Auth.js**; engineered type-safe server actions and Zod-validated schemas over a production-style relational backend.
- Implemented **end-to-end commerce flows**: credentials + Google OAuth, OTP verification, JWT sessions, RBAC, Razorpay checkout, server-side payment verification, webhook handling, ticket generation, and automated wallet rewards.
- Built a **rich admin suite** with a TipTap CMS editor enhanced by **Gemini AI** for writing assistance and SEO optimization, plus browser-based mini-games (Sudoku, Tetris, 2048) with custom game logic and scoring algorithms.

DesertFormer | Off-Road Semantic Segmentation

Python, PyTorch, Transformers

- Built an **11-class segmentation pipeline** in Python using **SegFormer/MiT backbones** on 4,176 images with rare-class-aware training (class-weighted 0.7 CE + 0.3 Dice loss), copy-paste augmentation, mixed-precision, and SWA.
- Achieved **66.5% val mIoU** and **87.7% pixel accuracy** with multi-scale TTA and optional DenseCRF; per-class IoU up to **98.1%** on sky; conducted detailed per-class performance analysis and iterative model debugging to improve rare-class detection.

Image Colorization using Pix2Pix (Conditional GANs)

[GitHub](#)

Implemented a **Pix2Pix cGAN** in Python for grayscale-to-RGB translation in PyTorch with a custom U-Net generator and PatchGAN discriminator; optimized with perceptual loss for high structural similarity and color fidelity.

Legal Document Summarization — JUST-NLP 2025 | IIT Kharagpur

[Competition Link](#)

Developed hybrid BART-based abstractive summarization models in Python for the **IIT Kharagpur CNeRG competition**; evaluated using ROUGE and BERTScore metrics, achieving top-tier performance on long-form legal clauses.

Kaggle Datathon — GDGC SVNIT (Rank 4 / 100+ teams)

[Event Page](#)

Built end-to-end ML pipelines in Python with XGBoost, CatBoost, and deep MLP ensembles; applied feature engineering, data cleaning, and hyperparameter optimization to achieve 4th-place finish across 100+ competing teams.

EDUCATION

2024 – 2028	B.Tech in Artificial Intelligence , SVNIT Surat	(CGPA: 7.78/10)
	<i>Relevant Coursework:</i> Data Structures & Algorithms, Automata & Formal Languages, Computer Networks, Database Systems, Artificial Intelligence, Knowledge Representation	
2023	JEE Main	(AIR 9082)
2023	Class XII — JAC Board	(78.8%)
2021	Class X — CBSE Board	(94%)

POSITIONS OF RESPONSIBILITY

- **Executive Member**, ACM SVNIT Chapter
- **Member**, Cutting Edge Visionaries (CEV)
- **Active Member**, GDGC SVNIT
- **Member**, Kirdar — Dramatics Club, SVNIT

SKILLS

Languages	Python, C++, C, TypeScript, JavaScript, SQL, HTML/CSS
Algorithms & DSA	Data Structures, Dynamic Programming, Graph Algorithms, Trees, Binary Search, Greedy Algorithms; Codeforces Specialist (rating 1554) , active on NeetCode (150+ problems)
Backend & Systems	Go, NestJS, Node.js, Express, Prisma, PostgreSQL, Redis, Socket.IO, WebSockets, REST APIs, JWT, Docker, Nginx
Frontend	Next.js, React, Tailwind CSS, Zustand, TanStack Query, Recharts, PWA
ML / AI	PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, Albumentations, OpenCV, GANs, NLP